Financial Modeling Projects Summary

2025 Introduction to Quantitative Methods in Finance
The Erdös Institute

Yichen Li

June 2025

Introduction

- Four mini projects on financial modeling and risk management
- Topics include portfolios, normality testing, option Greeks, and delta hedging
- Goal: understand practical implications of financial models

Project 1: Portfolio Volatility

- Constructed high- and low-volatility portfolios
- Used daily stock data from June 2023 to June 2024
- Compared returns and risks between the two portfolios

Table: Descriptive statistics of selected stocks (June 2023 – June 2024).

Ticker	Mean	Std	Min	Max	CI Lower	CI Upper
AAPL	181.05	8.36	164.01	196.67	180.02	182.08
AMZN	151.38	21.21	119.57	189.50	148.76	154.00
GOOGL	138.67	14.19	115.76	176.79	136.92	140.43
JNJ	149.89	5.75	137.72	163.85	149.18	150.60
JPM	159.32	20.96	130.99	200.15	156.73	161.90
MSFT	365.66	38.00	308.01	427.24	360.97	370.35
NVDA	59.59	20.17	37.45	114.78	57.10	62.09
TSLA	221.39	36.66	142.05	293.34	216.86	225.91
V	252.54	19.32	219.74	287.74	250.16	254.93
WMT	54.31	3.95	47.96	65.09	53.82	54.80

Project 1 Results

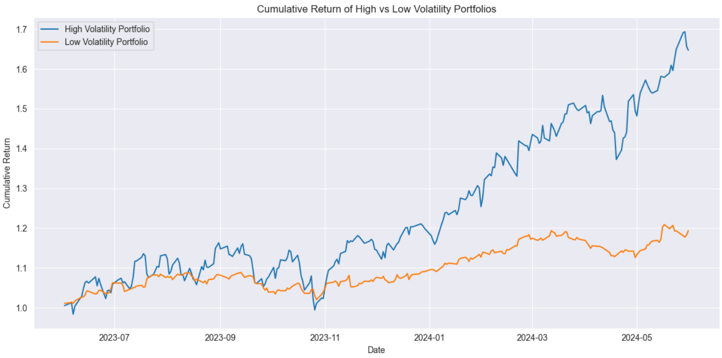


Figure: Cumulative return of high- vs low-volatility portfolios over one year.

- High-volatility portfolio: higher return, higher risk
- Low-volatility portfolio: lower but steadier return
- Demonstrates risk-return tradeoff in portfolio construction

Project 2: Return Normality

- Tested if portfolio returns follow a normal distribution
- Used histograms, QQ-plots, and statistical tests
- Important for risk management assumptions

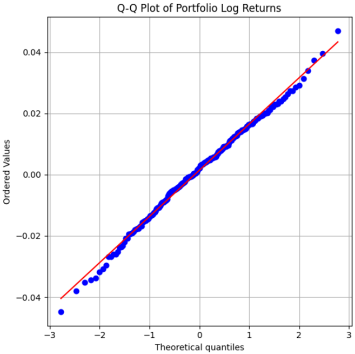


Figure: Normal Q–Q plot of high-volatility portfolio daily returns.

Project 2 Results

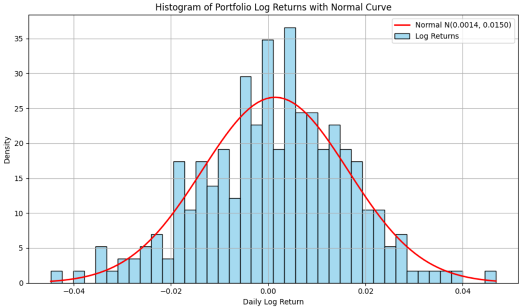


Figure: Histogram of portfolio daily log returns overlaid with fitted normal distribution curve.

- Portfolio returns approximately normal
- Minor deviations in tails (slightly heavier tails)
- Highlights need to consider extreme events in modeling

Project 3: Option Greeks

- Analyzed option sensitivities: Delta, Gamma, Vega
- Visualized Greeks across stock prices
- Studied time decay (Theta) effect on options

Figure: Black–Scholes option prices vs. spot price ($t = 0.5$ year).

Project 3 Results

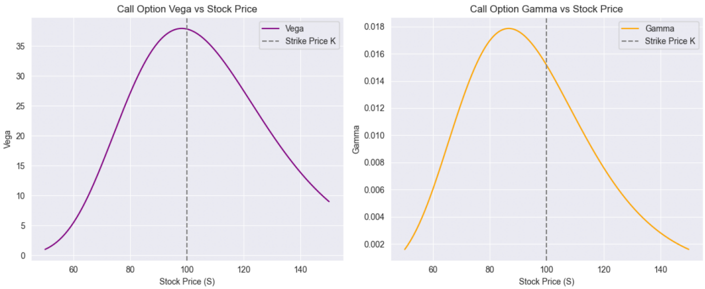


Figure: Option Vega (purple, left) and Gamma (orange, right) as functions of stock price (Black–Scholes, K=100, $t = 1$).

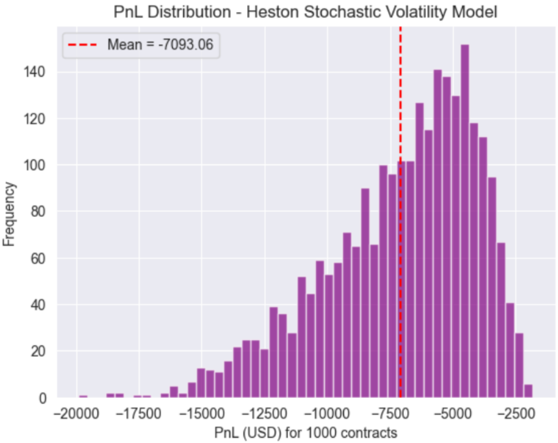
- Delta increases with stock price
- Gamma and Vega peak at-the-money
- Theta shows faster decay near expiration

Project 4: Delta Hedging

- Evaluated delta hedging under stochastic volatility
- Models: Regime Switching, Heston, GARCH
- Simulated PnL distributions under each model

Project 4 Results

- Hedging errors significant under stochastic volatility
- Heavy-tailed PnL distributions observed
- Delta hedging alone insufficient in volatile markets



`Mean: -7093.06, Std: 3025.16, Min: -19901.55, Max: -1830.82`

Comparative Analysis

- Projects show models have assumptions and limits
- High returns often come with higher risks
- Volatility modeling crucial for accurate hedging

Key Conclusions

- Financial models provide useful insights
- Model risk and tail risk remain important concerns
- Combining models and empirical analysis improves decisions

Personal Reflection

- Learned practical coding and modeling skills
- Improved understanding of financial risk management
- Gained experience with Python, LaTeX, and data analysis

Thank You!